

Multiple Choice (5 marks)

1. For the implication, $P \rightarrow Q$, the negation can be written as.

- A. $Q \rightarrow P$
- B. $\neg Q \rightarrow \neg P$
- C. $P \wedge \neg Q$
- D. None of the above

2. Which of the following functions provides a counterexample to the following proposition?

“If $f'(x) > 0$ for all $x \in \mathbb{R}$, then $f(x) > 0$ for all $x \in \mathbb{R}$.”

- A. $f(x) = x^2 + 1$
- B. $f(x) = x^2 - 1$
- C. $f(x) = 1 - x$
- D. $f(x) = x^3 + x + 1$

3. Which of the following is the negation of the statement “all warlords are dishonourable people”?

- A. If a person is honourable, they are not a warlord.
- B. Some people are either dishonourable or they are warlords.
- C. Some warlords are honourable people
- D. A person is honourable if they're not a warlord.

4. The value of $\int \frac{1}{(\sin \theta)^2 (\cos \theta)^2} d\theta$ is

- A. $\tan \theta + \cot \theta + c$
- B. $\tan \theta \cot \theta + c$
- C. $\tan \theta - \cot \theta + c$
- D. $\tan \theta - \cot 2\theta + c$

5. The value of $\int \frac{dx}{x(x^2+1)}$ is
- A. $\ln|x| + \tan^{-1} x + c$
 - B. $\ln|x| + \ln(1 + x^2) + c$
 - C. $\ln|x| - \tan^{-1} x + c$
 - D. $\ln|x| - \frac{1}{2}\ln(1 + x^2) + c$

Question 6 (15 marks)

a) Let S and P be the following propositions:

$S \equiv$ [I have studied for this examination]

$P \equiv$ [this examination is passable]

- i) Express $P \vee \neg S$ as an English sentence. (1 mark)
- ii) Write the statement given in (a) as an implication involving S and P, and express it also in English. (2 marks)

b) i) For all $a, b \in \mathbb{Z}$, show that 17 divides $10a+b$ if and only if 17 divides $a-5b$. (3 marks)

ii) Deduce, using part (i), that 17 divides 4080. (1 mark)

c) Using proof by contradiction, show that for all positive integers a, b and c,

if $a^2 + b^2 = c^2$, then either a is even or b is even. (3 marks)

d) i) Let $z=2+i$ and $\omega=1-2i$.

α) Find $z\omega$. (1 mark)

β) Find the angle between $z\omega$ and ω , leaving your answer to 2 decimal places. (2 mark)

ii) Let $z = re^{i\frac{\pi}{3}}$ where $r \in \mathbb{R}^+$ and $\omega = (3-3i)z$.

If w^k lies on a unit circle centred at the origin for all integer values of k , find the value of r . (2 marks)

Question 7 (start a new page) (15 marks)

a) i) Find a, b and c such that $\int x^2 e^{2x} dx = (ax^2 + bx + c)e^{2x}$. (2marks)

ii) Show that $\int_0^{\pi} \frac{x \sin x}{1 + (\cos x)^2} dx = \frac{\pi^2}{4}$. (2marks)

iii) Evaluate $\int \frac{\sin x}{\sin(x+a)} dx$, $a > 0$. (2marks)

(iv) Find $\frac{d}{dx} \left(\int_0^{\cos ax} \sqrt{1-t^2} dt \right)$. (2marks)

b) i) Show by taking the contrapositive, that for all positive integers n , n is even if n^2 is

even. (3 marks)

ii) Hence, using proof by contradiction, show that $\sqrt{8n-2}$ is irrational. (4 marks)

Question 8 (Start a new Page) (15 marks)

a) $5 - 2i$ is a solution to the equation $z^2 + az + b = 0$ where $a, b \in \mathbb{R}$. Find the value of a and b . (2 marks)

b) Let $z = 2(\cos 2\theta + i \sin 2\theta)$ where $0 < \theta < 45^\circ$. Find the modulus and argument of $z + 2$, expressing your answers in terms of θ . (2 marks)

c) Consider the complex number $z = \frac{\omega_1}{\omega_2}$ where $\omega_1 = \sqrt{2} + \sqrt{6}i$ and $\omega_2 = 3 + \sqrt{3}i$.

i) Express z in modulus-argument form. (3 marks)

ii) Find the smallest positive integer value of n such that z^n is a real number. (1 mark)

d) Sketch the region in the complex plane where the following inequalities hold simultaneously. (3 marks)

$$-\frac{\pi}{4} < \text{Arg}(z - 2 + 3i) < \frac{\pi}{4} \text{ and } |z - 2 + 3i| \leq 2$$

(Continue with Q8)

e) For the equation $z^3 + 64 = 0$, $z \in \mathbb{C}$.

i) Find the three distinct roots of the equation in modulus-argument form. (3 marks)

ii) The roots of the equation represent the vertices of a triangle. Find the area of the triangle formed. (1 mark)

Question 9 (Start a New Page) (15 marks)

a) i) $\int \frac{dx}{\cos \alpha + \cos x}$, $0 < \alpha < \pi$ (3 marks)

ii) $\int \frac{3x-2}{(x+1)^2(x+3)} dx$ (3 marks)

b) Consider the complex numbers $z_1 = -1 + i$ and $z_2 = \frac{1}{\sqrt{2}}(\cos(\frac{\pi}{3}) + i\sin(\frac{\pi}{3}))$.

i) Write $z_1 z_2$ in the form $a + ib$ where $a, b \in \mathbb{R}$. (2marks)

ii) Hence or otherwise, find the value of $\tan(\frac{\pi}{12})$ in the form $c + d\sqrt{3}$ where $c, d \in \mathbb{Z}$.

(3marks)

c) Given that $z^5 - 1 = (z - 1)(z^4 + z^3 + z^2 + z + 1)$.

Let ω be the complex root with the smallest positive argument to the solution to

$$z^5 - 1 = 0 \text{ where } \omega \in \mathbb{C}.$$

i) Prove that $1 + \omega^2 + \omega^4 = -(\omega + \omega^3)$ (1mark)

ii) Hence show that $\cos(\frac{2\pi}{5}) - \cos(\frac{\pi}{5}) = -\frac{1}{2}$ (3marks)

Question 10 (Start a New Page) (15 marks)

a) Use integration, find the area of the region in the first quadrant enclosed by the x-axis,

the line $y = x$ and the circle $x^2 + y^2 = 32$. (3 marks)

b) Given that $I_n = \int (x^2 - a^2)^{\frac{2n-1}{2}} dx$

i) Differentiate $x + \sqrt{x^2 - a^2}$, (1mark)

ii) Hence show that $I_0 = \ln |x + \sqrt{x^2 - a^2}|$ (2 marks)

iii) Use integration by parts to show that $I_n = \frac{1}{2n} x(x^2 - a^2)^{\frac{2n-1}{2}} - \frac{2n-1}{2n} a^2 I_{n-1}$

(3 marks)

c) Given that f is an integrable function over the closed interval $[0, a]$.

i) Prove that $\int_0^a f(x) dx = \int_0^{\frac{a}{2}} f(x) dx + \int_0^{\frac{a}{2}} f(a - x) dx$ (2 marks)

ii) Hence, or otherwise show that $\int_0^{\pi} \ln(\sin \theta) d\theta = -\pi \ln 2$. (4 marks)

END of PAPER

Ext 2 2021 Yr 12 Task 2

MATHEMATICS Extension 2: Question.....MC.....

Suggested Solutions

Marks

Marker's Comments

1. D

2. D

3. C

4. C

$$\begin{aligned}\int \frac{1}{\sin^2 \theta \cos^2 \theta} d\theta &= \int \frac{\sec^2 \theta}{\sin^2 \theta} d\theta \\ &= \int \frac{\tan^2 \theta + 1}{\sin^2 \theta} d\theta \\ &= \int \left(\frac{\frac{\sin^2 \theta}{\cos^2 \theta}}{\sin^2 \theta} + \frac{1}{\sin^2 \theta} \right) d\theta \\ &= \int \sec^2 \theta + \csc^2 \theta d\theta \\ &= \tan \theta - \cot \theta + C\end{aligned}$$

5. D

$$\frac{1}{x(x^2+1)} = \frac{A}{x} + \frac{Bx+C}{x^2+1}$$

$$1 = A(x^2+1) + (Bx+C)(x)$$

$$\text{when } x=0, A=1$$

$$x=1, 1 = 2 + B + C$$

$$x=-1, 1 = 2 + B - C$$

$$\therefore 2 = 4 + 2B$$

$$B = -1$$

$$\therefore C = 0$$

$$\begin{aligned}\therefore \int \frac{1}{x(x^2+1)} dx &= \int \left(\frac{1}{x} - \frac{x}{x^2+1} \right) dx \\ &= \ln|x| - \frac{1}{2} \ln(x^2+1) + C\end{aligned}$$

MATHEMATICS Extension 2: Question...6....

Suggested Solutions

Marks

Marker's Comments

a)

i I have not studied for this examination

①

ii $S \rightarrow P$

①

If I have studied for this examination,
then this examination is passable

①

It must be an
"if, then" meaning!

NOT "S happened,
therefore P".

b) i let $10a + b = 17k_1$, where $k_1 \in \mathbb{Z}$

$$\therefore b = 17M_1 - 10a$$

$$\therefore a - 5b = a - 5(17k_1 - 10a)$$

$$= a - 85k_1 + 50a$$

$$= 51a - 85k_1$$

$$= 17(3a - 5k_1)$$

$$= 17M_1 \text{ where } M_1 \in \mathbb{Z} \\ (\text{Since } a, k_1 \in \mathbb{Z})$$

① for attempting
to prove the implication
from both directions

① for correctly proving
first implication

① for correctly proving
second implication

let $a - 5b = 17k_2$ where $k_2 \in \mathbb{Z}$,

$$\therefore a = 17k_2 + 5b$$

$$\therefore 10a + b = 10(17k_2 + 5b) + b$$

$$= 170k_2 + 50b + b$$

MATHEMATICS Extension 2: Question.....

Suggested Solutions

Marks

Marker's Comments

$$= 170k_2 + 51b$$

$$= 17(10k_2 + 3b)$$

$$= 17M_2 \text{ where } M_2 \in \mathbb{Z} \\ (\text{since } k_2, b \in \mathbb{Z})$$

$\therefore 17$ divides $10a+b$ iff 17 divides $a-5b$

See next page for b) ii

c) Assume that $a^2 + b^2 = c^2$ where $a, b, c \in \mathbb{Z}$ and a, b are odd

$$\text{let } a = 2k_1 + 1 \quad b = 2k_2 + 1$$

$$\therefore c^2 = a^2 + b^2$$

$$= (2k_1 + 1)^2 + (2k_2 + 1)^2$$

$$= (4k_1^2 + 4k_1 + 1) + (4k_2^2 + 4k_2 + 1)$$

$$= 2(2k_1^2 + 2k_1 + 2k_2^2 + 2k_2 + 1)$$

$\therefore c^2$ is even

$\therefore c$ is even (property of an even number)

$$\therefore c = 2p$$

$$\therefore 4p^2 = 2(2k_1^2 + 2k_1 + 2k_2^2 + 2k_2 + 1)$$

$$2p^2 = 2k_1^2 + 2k_1 + 2k_2^2 + 2k_2 + 1$$

LHS is even, RHS = odd

$\therefore$ original statement is proven by contradiction.

① Students Do NOT get this mark if they attempt to assume both a, b are even and both a, b are odd

①

①

MATHEMATICS Extension 2: Question.....

Suggested Solutions

Marks

Marker's Comments

b) i) let $4080 = 10 \times 408 + 0$

$\therefore a = 408$ and $b = 0$ for $10a + b$

$a - 5b = 408$

$408 \div 17 = 24$

$\therefore 17$ divides 408

$\therefore 17$ divides 4080 (from part i)) — ①

MATHEMATICS Extension 2: Question.....

Suggested Solutions

Marks

Marker's Comments

d) i)

$$d) \quad zw = (2+i)(1-2i)$$

$$= 2 - 4i + i + 2$$

$$= 4 - 3i$$

①

$$b) \quad \arg(zw) - \arg(w)$$

$$= \arg\left(\frac{zw}{w}\right)$$

$$= \arg(z)$$

①

$$= \tan^{-1}\left(\frac{1}{2}\right)$$

$$= 26.57^\circ \text{ or } 0.46 \text{ rad}$$

①

$$ii) \quad |w^k| = 1$$

$$\therefore |w| = 1$$

①

$$\therefore |(3-3i)z| = 1$$

$$|3-3i| |z| = 1$$

$$\sqrt{18} |z| = 1$$

$$\therefore |z| = \frac{1}{\sqrt{18}}$$

$$= \frac{1}{3\sqrt{2}}$$

$$\therefore r = \frac{1}{3\sqrt{2}} \text{ or } \frac{\sqrt{2}}{6}$$

①

Suggested Solutions

Marks

Marker's Comments

ai) Method 1

$$\frac{d}{dx}(ax^2+bx+c)e^{2x} = (2ax+b)e^{2x} + 2(ax^2+bx+c)e^{2x}$$

$$= (2ax^2 + 2(a+b)x + (b+2c))e^{2x}$$

Match with x^2e^{2x}

$$\begin{cases} 2a=1 \\ 2(a+b)=0 \\ b+2c=0 \end{cases} \Rightarrow \begin{cases} a=\frac{1}{2} \\ b=-\frac{1}{2} \\ c=\frac{1}{4} \end{cases}$$

Method 2

$$u=x^2, \frac{dv}{dx}=e^{2x}$$

$$du=2xdx, v=\frac{1}{2}e^{2x}$$

$$\int x^2 e^{2x} dx = \frac{1}{2} x^2 e^{2x} - \int \frac{1}{2} e^{2x} 2x dx$$

$$= \frac{1}{2} x^2 e^{2x} - \int e^{2x} x dx$$

$$u=x, \frac{dv}{dx}=e^{2x}$$

$$du=dx, v=\frac{1}{2}e^{2x}$$

$$\int x^2 e^{2x} dx = \frac{1}{2} x^2 e^{2x} - \left(\frac{1}{2} x e^{2x} - \int \frac{1}{2} e^{2x} dx \right)$$

$$= \frac{1}{2} x^2 e^{2x} - \frac{1}{2} x e^{2x} + \frac{1}{2} \int e^{2x} dx$$

$$= \frac{1}{2} x^2 e^{2x} - \frac{1}{2} x e^{2x} + \frac{1}{4} e^{2x} + C_1$$

$$= \left(\frac{1}{2} x^2 - \frac{1}{2} x + \frac{1}{4} \right) e^{2x} + \textcircled{C_1}$$

Match with $(ax^2+bx+c)e^{2x}$

$$a=\frac{1}{2}, b=-\frac{1}{2}, c=\frac{1}{4}$$

↓
constant term,
has nothing to do
with e^{2x}

The second integration
by parts was
done poorly.

Many students
forgot to write
down the values
for a, b, and c.

Suggested Solutions

Marks

Marker's Comments

$$a ii) \int_0^{\pi} \frac{x \sin x}{1 + (\cos x)^2} dx$$

$$= \int_0^{\pi} \frac{(\pi - x) \sin(\pi - x)}{1 + \cos^2(\pi - x)} dx \quad (\text{dummy variables}).$$

$$= \int_0^{\pi} \frac{(\pi - x) \sin x}{1 + \cos^2 x} dx$$

$$= \int_0^{\pi} \frac{\pi \sin x}{1 + \cos^2 x} dx - \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx$$

$$2 \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx = \int_0^{\pi} \frac{\pi \sin x}{1 + \cos^2 x} dx$$

$$= -\pi \int_0^{\pi} \frac{-\sin x}{1 + \cos^2 x} dx$$

$$-\sin x = \frac{d(\cos x)}{dx}$$

$$= -\pi \int_0^{\pi} \tan^{-1}(\cos x) dx$$

$$= -\pi \left[-\frac{\pi}{4} - \frac{\pi}{4} \right] = \frac{\pi^2}{2}$$

$$\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx = \frac{\pi^2}{4}$$

1

Some students
used $\frac{\pi}{2} - x$
instead of $\pi - x$

1

Suggested Solutions

Marks

Marker's Comments

 ciii) Method 1 *The goal is to make the denominator simply.*

$$\int \frac{\sin x}{\sin(x+a)} dx$$

$$\text{let } u = x+a, \quad x = u-a, \\ du = dx$$

$$= \int \frac{\sin(u-a)}{\sin u} du$$

$$= \int \frac{\sin u \cos a - \sin a \cos u}{\sin u} du$$

$$= \int \cos a - \sin a \frac{\cos u}{\sin u} du$$

$$= (\cos a)u - \sin a \ln|\sin u| + C$$

$$= (\cos a)(x+a) - \sin a \ln|\sin(x+a)| + C$$

Method 2

$$\int \frac{\sin x}{\sin(x+a)} dx = \int \frac{\sin[(x+a)-a]}{\sin(x+a)} dx$$

$$= \int \frac{\sin(x+a)\cos a - \cos(x+a)\sin a}{\sin(x+a)} dx$$

$$= \int \cos a dx - \sin a \int \frac{\cos(x+a)}{\sin(x+a)} dx$$

$$= (\cos a)x - \sin a \ln|\sin(x+a)| + C$$

 Some students substitute $u = \sin(x+a)$ but it doesn't make it simply

 cannot use dummy variable to swap it back to $\int \frac{\sin(x-a)}{\sin x} dx$

 Some students forgot to change back to x .

Suggested Solutions

Marks

Marker's Comments

aiv) Method 1

Method 1 should not have calculation of lower bound 0.

using $\frac{d}{dx} \int_a^x f(t) dt = f(x)$

let $f(t) = \sqrt{1-t^2}$

$$I = \frac{d}{d(\cos ax)} \int_0^{\cos ax} f(t) dt \times \frac{d \cos ax}{dx}$$

$$= f(\cos ax) \times (-a \sin ax)$$

$$= \sqrt{1-\cos^2 ax} \times (-a \sin ax)$$

$$= -a \sin^2 ax \quad (0 \leq x \leq \frac{\pi}{2a}, 0 \leq \sin ax \leq 1)$$

$$\sqrt{1-\cos^2 ax} = |\sin ax|$$

Method 2

Using $\int_a^b f(x) dx = F(b) - F(a)$

let $f(t) = \sqrt{1-t^2}$, $F(t) = \int \sqrt{1-t^2} dt$

$$\frac{d}{dx} \int_0^{\cos ax} f(t) dt$$

$$= \frac{d}{dx} [F(\cos ax) - F(0)]$$

$$= f(\cos ax) \frac{d \cos ax}{dx} - 0$$

$$= \sqrt{1-\cos^2 ax} \cdot (-a \sin ax)$$

$$= -a \sin^2(ax) \quad (0 \leq x \leq \frac{\pi}{2a}, 0 \leq \sin ax \leq 1)$$

 showing $f(\cos ax)$ or equivalent form and.

 acknowledgement of $\frac{d \cos ax}{dx}$ in

certain form,

 such as $\frac{f(\cos ax)}{a}$

 or $\frac{f(\cos ax)}{a} (-\sin ax)$

receives 1 mark

 students need to define $F(t)$ and $f(t)$ if they use it during the calculation.

Bound calculation or its equivalent form is required for this method.

Suggested Solutions	Marks	Marker's Comments
bi) RTP: n^2 is even $\Rightarrow n$ is even. Contrapositive: n is odd $\Rightarrow n^2$ is odd If n is odd, let $n = 2k-1$, $k \in \mathbb{Z}^+$ $n^2 = (2k-1)^2$ $= 4k^2 - 4k + 1$ $= 2(2k^2 - 2k) + 1$ $\therefore 2k^2 - 2k \in \mathbb{Z} \rightarrow$ many students didn't write this $\therefore n^2$ is odd. $\therefore$ contrapositive is proven. $\therefore$ the original statement is true.	1 1 1	many students get the order wrong. n is a positive integer, $n = 2k+1$, $k \in \mathbb{Z}$ doesn't guarantee that.

Suggested Solutions

Marks

Marker's Comments

 bii) Suppose $\sqrt{8n-2} \in \mathbb{Q}$

$$\text{let } \sqrt{8n-2} = \frac{p}{q_n} \quad (p, q_n \in \mathbb{Z}, q_n \neq 0, \\ p, q_n \text{ coprime})$$

$$(8n-2) = \frac{p^2}{q_n^2}$$

$$2(4n-1)q_n^2 = p^2 \quad \textcircled{1}$$

$$\therefore (4n-1)q_n^2 \in \mathbb{Z}$$

$$\therefore p^2 \text{ is even.}$$

$$\therefore p \text{ is even (by b(i) statement)}$$

$$\text{let } p = 2k \quad (k \in \mathbb{Z})$$

→ don't use the same letter that the question already used.

$$\text{from } \textcircled{1} \quad (8n-2)q_n^2 = 4k^2$$

$$(4n-1)q_n^2 = 2k^2$$

$$\therefore 4n-1 \text{ is odd}$$

$$2k^2 \text{ is even}$$

$$\therefore q_n^2 \text{ is even}$$

$$\therefore q_n \text{ is even (by b(i) statement)}$$

$\therefore$ both p and q_n are both even,
 which means that they aren't coprime.
 which is contradict with the initial
 assumption that p and q_n are coprime.

$$\therefore \sqrt{8n-2} \text{ is irrational}$$

1

Some students use

$$2(4n-1) = \frac{p^2}{q_n^2}$$

$$\frac{p^2}{q_n^2} \text{ is even} \Rightarrow \frac{p}{q_n}$$

is even

bi) only applies
 to positive integers
 not including fractions

1

many students
 did not write
 this part.

common mistake:

if two numbers
 are divisible by
 the same number,
 they are coprime.

1

MATHEMATICS Extension 2: Question...8...

Suggested Solutions

Marks

Marker's Comments

a) Since $5-2i$ is a root, it will satisfy the equation.

$$\therefore (5-2i)^2 + a(5-2i) + b = 0$$

$$25 - 20i - 4 + 5a - 2ai + b = 0$$

$$21 + 5a + b - (20 + 2a)i = 0$$

equating real and imaginary parts.

$$\therefore 20 + 2a = 0$$

$$a = -10$$

$$\therefore 21 + 5a + b = 0$$

$$21 + 5(-10) + b = 0$$

$$b = 29$$

$$\therefore a = -10, b = 29$$

OR since $5-2i$ is a root, then $5+2i$ is also a root as coefficients are real.

sum of roots:

$$5-2i + 5+2i = -a$$

$$\therefore a = -10$$

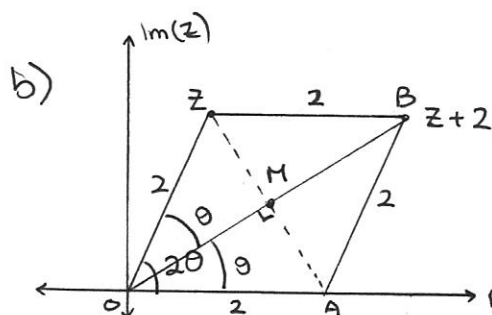
product of roots:

$$(5-2i)(5+2i) = b$$

$$b = 25 + 4$$

$$b = 29$$

$$\therefore a = -10, b = 29$$



OABZ is a rhombus
with a pair of adjacent
sides equal and opposite
sides equal and parallel.

MATHEMATICS Extension 2: Question...8...

Suggested Solutions	Marks	Marker's Comments
$\therefore \arg(z+2) = \frac{1}{2} \arg z$ $= \theta$ In $\triangle OMA$, $\cos \theta = \frac{OM}{2}$ $OM = 2 \cos \theta$ $ z+2 = 2(OM)$ $= 2(2 \cos \theta)$ $= 4 \cos \theta.$ OR $z+2 = 2(\cos 2\theta + i \sin 2\theta) + 2$ $= 2(2 \cos^2 \theta - 1 + 2i \sin \theta \cos \theta) + 2$ $= 4 \cos^2 \theta + 4i \sin \theta \cos \theta$ $= 4 \cos \theta (\cos \theta + i \sin \theta)$ $\therefore z+2 = 4 \cos \theta$ $\arg(z+2) = \theta$ OR $z+2 = 2 \cos 2\theta + 2 + 2i \sin 2\theta$ $ z+2 = \sqrt{(2 \cos 2\theta + 2)^2 + (2 \sin 2\theta)^2}$ $= \sqrt{4 \cos^2 2\theta + 8 \cos 2\theta + 4 + 4 \sin^2 2\theta}$ $= \sqrt{4 + 8 \cos 2\theta + 4}$ $= \sqrt{8 + 8(2 \cos^2 \theta - 1)}$ $= \sqrt{16 \cos^2 \theta}$ $= 4 \cos \theta$ $\arg(z+2) = \tan^{-1} \left(\frac{2 \sin 2\theta}{2 \cos 2\theta + 2} \right)$ $= \tan^{-1} \left(\frac{2 \sin \theta \cos \theta}{2 \cos^2 \theta - 1 + 1} \right)$ $= \tan^{-1} \left(\frac{\sin \theta}{\cos \theta} \right)$ $= \tan^{-1} (\tan \theta)$ $= \theta$	1 1 1 1 1 1	A lot of students forgot the square root for the modulus.

MATHEMATICS Extension 2: Question...8...

Suggested Solutions	Marks	Marker's Comments
$c) i) z = \frac{\sqrt{2} + \sqrt{6}i}{3 + \sqrt{3}i}$ $= \frac{2\sqrt{2} \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)}{2\sqrt{3} \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right)}$ $= \frac{\sqrt{2}}{\sqrt{3}} \left(\cos \left(\frac{\pi}{3} - \frac{\pi}{6} \right) + i \sin \left(\frac{\pi}{3} - \frac{\pi}{6} \right) \right)$ $= \frac{\sqrt{6}}{3} \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right)$ OR $z = \frac{\sqrt{2} + \sqrt{6}i}{3 + \sqrt{3}i} \times \frac{3 - \sqrt{3}i}{3 - \sqrt{3}i}$ $= \frac{3\sqrt{2} - \sqrt{6}i + 3\sqrt{6}i + \sqrt{18}}{9 + 3}$ $= \frac{3\sqrt{2} + 2\sqrt{6}i + 3\sqrt{2}}{12}$ $= \frac{6\sqrt{2} + 2\sqrt{6}i}{12}$ $= \frac{\sqrt{2}}{2} + \frac{\sqrt{6}}{6}i$ $\therefore z = \sqrt{\left(\frac{\sqrt{2}}{2} \right)^2 + \left(\frac{\sqrt{6}}{6} \right)^2}$ $= \sqrt{\frac{2}{3}}$ $\arg(z) = \tan^{-1} \left(\frac{\frac{\sqrt{6}}{6}}{\frac{\sqrt{2}}{2}} \right)$ $= \tan^{-1} \left(\frac{1}{\sqrt{3}} \right)$ $= \frac{\pi}{6}$ $\therefore z = \sqrt{\frac{2}{3}} \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right)$	1 1 1 1 1 1 1	

MATHEMATICS Extension 2: Question...8...

Suggested Solutions

Marks

Marker's Comments

ii) for z^n to be real, $\text{Im}(z^n) = 0$
 $z^n = \left(\frac{\sqrt{6}}{3}\right)^n \left(\cos \frac{n\pi}{6} + i \sin \frac{n\pi}{6}\right)$ (De Moivre's Theorem)

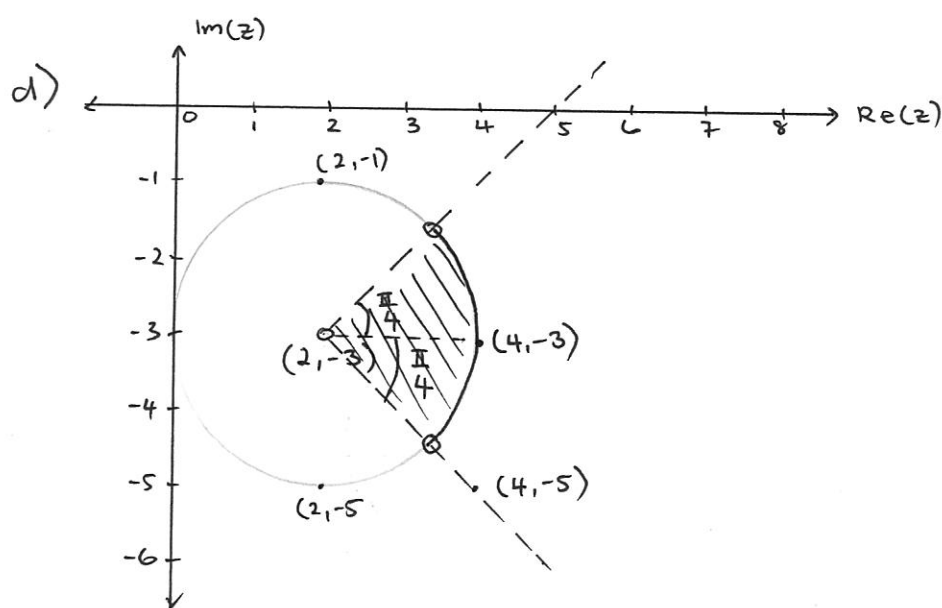
$\therefore \sin \frac{n\pi}{6} = 0$

$\frac{n\pi}{6} = 0, \pi, 2\pi, \dots$

$n = 0, 6, 12, \dots$

$\therefore$ smallest positive integer is 6.

1



1

circle - centre

- radius

- solid line

1

lines - open circle at (2, -3)

- dotted line

- indicate angle or correct point indicated on each line

1

correct shaded area.

e) $z^3 + 64 = 0$

i) $z^3 = -64$

let $z = r(\cos \theta + i \sin \theta)$

$z^3 = r^3(\cos 3\theta + i \sin 3\theta)$ (De Moivre's Theorem)

$\therefore r^3(\cos 3\theta + i \sin 3\theta) = 64(\cos \pi + i \sin \pi)$

1

$r^3 = 64$

$3\theta = \pi + 2k\pi$ for $k \in \mathbb{Z}$

$\therefore r = 4$

$\therefore \theta = \frac{\pi + 2k\pi}{3}$

1

$\therefore z = 4 \left(\cos \left(\frac{\pi + 2k\pi}{3} \right) + i \sin \left(\frac{\pi + 2k\pi}{3} \right) \right)$

MATHEMATICS Extension 2: Question....8...

Suggested Solutions

Marks

Marker's Comments

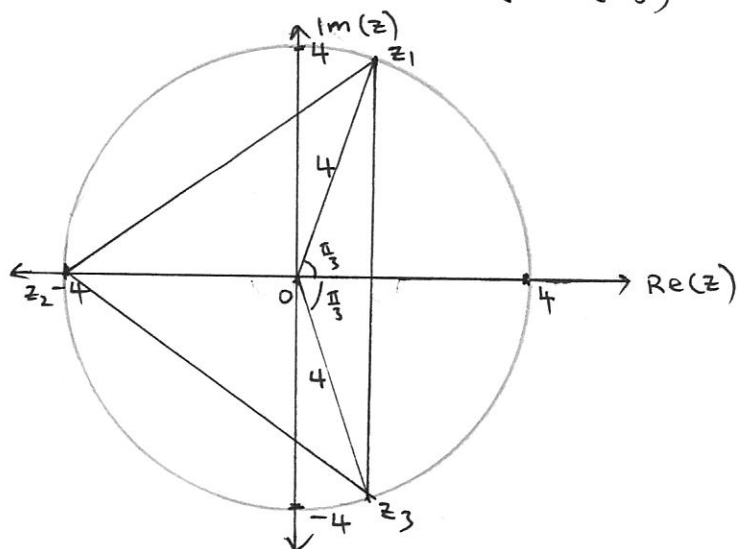
for $k=0$, $z_1 = 4(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})$

$k=1$, $z_2 = 4(\cos \pi + i \sin \pi)$
 $= -4$

$k=2$, $z_3 = 4(\cos(-\frac{\pi}{3}) + i \sin(-\frac{\pi}{3}))$

$\therefore$ the three roots are $4(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})$,
 $4(\cos \pi + i \sin \pi)$ and $4(\cos(-\frac{\pi}{3}) + i \sin(-\frac{\pi}{3}))$

ii)



triangle is made of three smaller triangles

$$\begin{aligned} \therefore \text{Area} &= 3 \times \frac{1}{2} \times 4 \times 4 \times \sin \frac{2\pi}{3} \\ &= 24 \times \frac{\sqrt{3}}{2} \\ &= 12\sqrt{3} \quad u^2 \end{aligned}$$

1

Question 9:

(a) (i) use the substitution $t = \tan \frac{x}{2}$,
find $\int \frac{dx}{\cos \alpha + \cos x}$, $0 < \alpha < \pi$

$$t = \tan \frac{x}{2} \quad \therefore dt = \frac{1}{2} (1 + \tan^2 \frac{x}{2}) dx$$

$$\frac{2 dt}{1+t^2} = dx, \quad \cos x = \frac{1-t^2}{1+t^2}$$

$$\int \frac{dx}{\cos \alpha + \cos x} = \int \frac{1}{\cos \alpha + \frac{1-t^2}{1+t^2}} \times \frac{2 dt}{1+t^2} \quad \checkmark$$

$$= \int \frac{2 dt}{(1+t^2) \cos \alpha + 1 - t^2}$$

$$= \int \frac{2 dt}{t^2 (\cos \alpha - 1) + 1 + \cos \alpha}$$

$$= \frac{2}{\cos \alpha - 1} \int \frac{dt}{t^2 - \frac{1 + \cos \alpha}{1 - \cos \alpha}} \quad \checkmark$$

$$1 + \cos \alpha = 2 \cos^2 \alpha/2 \quad \text{and} \quad 1 - \cos \alpha = 2 \sin^2 \alpha/2$$

$$\therefore \frac{1 + \cos \alpha}{1 - \cos \alpha} = \cot^2 \alpha/2$$

$$\int \frac{dx}{\cos \alpha + \cos x} = \frac{2}{\cos \alpha - 1} \int \frac{dt}{t^2 - \cot^2(\alpha/2)} = \frac{2}{\cos \alpha - 1} \left[\int \frac{\frac{2 \cot \alpha/2}{t - \cot \alpha/2} dt}{t - \cot \alpha/2} - \int \frac{\frac{2 \cot \alpha/2}{t + \cot \alpha/2} dt}{t + \cot \alpha/2} \right]$$

$$= \frac{2}{\cos \alpha - 1} \times \frac{1}{2 \cot \alpha/2} \left\{ \ln |t - \cot \alpha/2| - \ln |t + \cot \alpha/2| \right\} + C$$

$$= \frac{1}{\sin \alpha} \ln \left| \frac{t + \cot \alpha/2}{t - \cot \alpha/2} \right| + C \quad \checkmark$$

$$= \frac{1}{\sin \alpha} \ln \left| \frac{\tan x/2 + \cot \alpha/2}{\tan x/2 - \cot \alpha/2} \right| + C$$

$$(ii) \int \frac{3x-2}{(x+1)^4(x+3)} dx$$

$$\frac{3x-2}{(x+3)(x+1)^4} = \frac{A}{x+3} + \frac{B}{x+1} + \frac{C}{(x+1)^2}$$

multiply by $x+3$, then sub $x=-3$

$$A = \frac{3(-3)-2}{(-3+1)^2} = -\frac{11}{4}$$

Multiply by $(x+1)^4$, then sub $x=-1$

$$C = \frac{3(-1)-2}{(-1+3)} = -\frac{5}{2}$$

$$\text{Sub } x=0 \therefore \frac{-2}{3} = \frac{A}{3} + B + C$$

$$-2 = A + 3B + 3C$$

$$-2 = -\frac{11}{4} + 3B - \frac{15}{2}$$

$$8 = -11 + 12B - 30$$

$$33 = 12B \therefore B = \frac{33}{12} = \frac{11}{4}$$

$$\int \frac{3x-2}{(x+1)^4(x+3)} dx = -\frac{11}{4} \int \frac{dx}{x+3} + \frac{11}{4} \int \frac{dx}{x+1} - \frac{5}{2} \int \frac{dx}{(x+1)^2} \quad \checkmark$$

$$= -\frac{11}{4} \ln|x+3| + \frac{11}{4} \ln|x+1| + \frac{5}{2} \frac{1}{x+1} + C \quad \checkmark$$

$$= \frac{11}{4} \ln \left| \frac{x+1}{x+3} \right| + \frac{5}{2(x+1)} + C$$

$$(6) \quad Z_1 = -1 + i, \quad Z_2 = \frac{1}{\sqrt{2}} \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) \\ = \frac{1}{\sqrt{2}} \left(\frac{1}{2} + i \frac{\sqrt{3}}{2} \right) = \frac{\sqrt{2}}{4} + i \frac{\sqrt{6}}{4}$$

$$(i) \quad Z_1 Z_2 = (-1 + i) \left(\frac{\sqrt{2}}{4} + i \frac{\sqrt{6}}{4} \right) \quad \checkmark$$

$$= -\frac{\sqrt{2}}{4} - i \frac{\sqrt{6}}{4} + i \frac{\sqrt{2}}{4} - \frac{\sqrt{6}}{4}$$

$$= -\frac{(\sqrt{2} + \sqrt{6})}{4} - i \frac{(\sqrt{6} - \sqrt{2})}{4}$$

$$\checkmark \quad \frac{-1 - \sqrt{3}}{2\sqrt{2}} + \frac{1 - \sqrt{3}i}{2\sqrt{2}}$$

$$(ii) \quad Z_1 = -1 + i = \sqrt{2} \left(\frac{-1}{\sqrt{2}} + i \frac{1}{\sqrt{2}} \right) = \sqrt{2} \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right)$$

$$Z_1 Z_2 = \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right)$$

$$= \cos \left(\frac{\pi}{3} + \frac{3\pi}{4} \right) + i \sin \left(\frac{\pi}{3} + \frac{3\pi}{4} \right)$$

$$= \cos \frac{13\pi}{12} = \cos \left(\pi + \frac{\pi}{12} \right) \quad \checkmark$$

$$\cos \frac{13\pi}{12} = -\frac{(\sqrt{2} + \sqrt{6})}{4}, \quad \sin \frac{13\pi}{12} = \frac{\sqrt{2} - \sqrt{6}}{4}$$

$$\cos \frac{\pi}{12} = \frac{\sqrt{2} + \sqrt{6}}{4}, \quad \sin \frac{\pi}{12} = \frac{\sqrt{6} - \sqrt{2}}{4} \quad \checkmark$$

$$\tan \frac{\pi}{12} = \frac{\sqrt{6} - \sqrt{2}}{\sqrt{2} + \sqrt{6}} = \frac{\sqrt{3} - 1}{\sqrt{3} + 1} = \frac{(\sqrt{3} - 1)^2}{3 - 1} = \frac{4 - 2\sqrt{3}}{2} = 2 - \sqrt{3} \quad \checkmark$$

$$(C) \quad z^5 - 1 = (z-1)(z^4 + z^3 + z^2 + z + 1)$$

(i) ω is the non-real root $\therefore$

$$\omega^4 + \omega^3 + \omega^2 + \omega + 1 = 0 \therefore 1 + \omega^2 + \omega^4 = -\omega - \omega^3 = -(\omega + \omega^3) \quad (1)$$

(ii) divide by ω^2 , we obtain

$$\frac{1}{\omega^2} + 1 + \omega^2 = -\left(\frac{1}{\omega} + \omega\right)$$

$$z^5 - 1 = 0 \therefore z^5 = 1 = e^{0 + 2K\pi i}$$

$$\therefore z = e^{2K\pi i/5} \quad K = -2, -1, 0, 1, 2 \quad \checkmark$$

$$\frac{1}{\omega^2} + \omega^2 = 2 \cos 2\theta = 2 \cos \frac{4\pi}{5}, \quad \theta = \frac{2\pi}{5}$$

$$\frac{1}{\omega} + \omega = 2 \cos \theta = 2 \cos \frac{2\pi}{5}$$

$$\frac{1}{\omega^2} + \omega^2 = -2 \cos \frac{\pi}{5} \quad \checkmark$$

$$-2 \cos \frac{\pi}{5} + 1 = -2 \cos \left(\frac{2\pi}{5} \right)$$

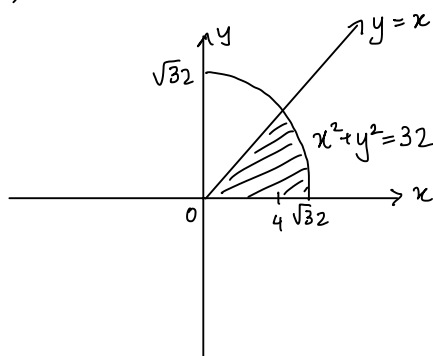
$$2 \cos \left(\frac{2\pi}{5} \right) - 2 \cos \frac{\pi}{5} = -1 \quad \checkmark$$

$$\cos \left(\frac{2\pi}{5} \right) - \cos \frac{\pi}{5} = -\frac{1}{2}$$

Question 10

Tuesday, 20 April 2021 11:45 am

a)



Find pt of intersection =

$$x^2 + y^2 = 32, y = x$$

$$\therefore 2x^2 = 32$$

$$x^2 = 16$$

$$x = 4, x > 0$$

$$y = \sqrt{32 - x^2}$$

Note: Sketch the diagram
Use integration
Check solution using
Must deal with $\sqrt{32}$
correctly for any
marks.

$$\text{Area} = \int_0^4 x \, dx + \int_4^{\sqrt{32}} \sqrt{32 - x^2} \, dx$$

$$= \left[\frac{x^2}{2} \right]_0^4 + \int_{\pi/4}^{\pi/2} \sqrt{32} \cos \theta \cdot \sqrt{32} \cos \theta \, d\theta$$

$$= \frac{16}{2} - 0 + 32 \int_{\pi/4}^{\pi/2} \cos^2 \theta \, d\theta$$

$$= 8 + 16 \int_{\pi/4}^{\pi/2} (1 + \cos 2\theta) \, d\theta$$

$$= 8 + 16 \left[\theta + \frac{\sin 2\theta}{2} \right]_{\pi/4}^{\pi/2}$$

$$= 8 + 16 \left[\frac{\pi}{2} + 0 - \left(\frac{\pi}{4} + \frac{1}{2} \right) \right]$$

$$= 8 + 16 \left[\frac{\pi}{4} - \frac{1}{2} \right]$$

$$= 8 + 4\pi - 8$$

$$= 4\pi \text{ square units}$$

① Integral expression for area

$$x = \sqrt{32} \sin \theta$$

$$\frac{dx}{d\theta} = \sqrt{32} \cos \theta$$

$$x = 4, \theta = \pi/4$$

$$x = \sqrt{32}, \theta = \pi/2$$

① double angle conversion

① correct numerical
expression for area

Question 10

Tuesday, 20 April 2021 11:58 am

$$b) (i) \frac{d}{dx} (x + \sqrt{x^2 - a^2})$$

$$= 1 + \frac{1}{2\sqrt{x^2 - a^2}} \times 2x$$

$$= 1 + \frac{x}{\sqrt{x^2 - a^2}} \quad (1)$$

$$(ii) I_0 = \int (x^2 - a^2)^{-1/2} dx$$

$$= \int \frac{1}{\sqrt{x^2 - a^2}} dx$$

$$= \int \frac{x + \sqrt{x^2 - a^2}}{\sqrt{x^2 - a^2} (x + \sqrt{x^2 - a^2})} dx \quad (1)$$

$$= \int \frac{\frac{x}{\sqrt{x^2 - a^2}} + 1}{x + \sqrt{x^2 - a^2}} dx \quad (1)$$

$$= \ln |x + \sqrt{x^2 - a^2}| + C$$

$$(iii) I_n = \int (x^2 - a^2)^{\frac{2n-1}{2}} dx$$

$$u = (x^2 - a^2)^{\frac{2n-1}{2}} \quad v' = 1$$

$$u' = \left(\frac{2n-1}{2}\right) (x^2 - a^2)^{\frac{2n-3}{2}} 2x \quad v = x$$

$$= (2n-1) (x^2 - a^2)^{\frac{2n-3}{2}} x$$

$$I_n = x (x^2 - a^2)^{\frac{2n-1}{2}} - \int (2n-1) x^2 (x^2 - a^2)^{\frac{2n-3}{2}} dx \quad (1)$$

$$= x (x^2 - a^2)^{\frac{2n-1}{2}} - (2n-1) \int (x^2 - a^2 + a^2) (x^2 - a^2)^{\frac{2n-3}{2}} dx \quad (1)$$

$$= x (x^2 - a^2)^{\frac{2n-1}{2}} - (2n-1) \int [(x^2 - a^2)^{\frac{2n-1}{2}} + a^2 (x^2 - a^2)^{\frac{2n-3}{2}}] dx$$

$$= x (x^2 - a^2)^{\frac{2n-1}{2}} - (2n-1) I_n - a^2 (2n-1) I_{n-1}$$

$$(2n-1) I_n + I_n = x (x^2 - a^2)^{\frac{2n-1}{2}} - a^2 (2n-1) I_{n-1} \quad (1)$$

$$I_n = \frac{1}{2n} x (x^2 - a^2)^{\frac{2n-1}{2}} - \frac{(2n-1)}{2n} a^2 I_{n-1}$$

Question 10

Tuesday, 20 April 2021 12:11 pm

$$c)(i) \int_0^a f(x) dx = \int_0^{a/2} f(x) dx + \int_{a/2}^a f(x) dx \quad \textcircled{i}$$

Consider $\int_{a/2}^a f(x) dx$

Let $x = a - u$

$$\int_{a/2}^a f(x) dx = \int_{a/2}^0 f(a-u) (-du)$$

$$\frac{dx}{du} = -1$$

$$dx = -du$$

$$= \int_0^{a/2} f(a-u) du$$

When $x = a/2$, $u = a/2$
 $x = a$, $u = 0$

$$= \int_0^{a/2} f(a-x) dx$$

dummy variable $\textcircled{i}$
(or substituting x for u)

$$\therefore \int_0^a f(x) dx = \int_0^{a/2} f(x) dx + \int_0^{a/2} f(a-x) dx$$

$$\begin{aligned}
 \text{ii, } \int_0^\pi \ln(\sin \theta) d\theta &= \int_0^{\pi/2} \ln(\sin \theta) d\theta + \int_{\pi/2}^\pi \ln(\sin(\pi - \theta)) d\theta \\
 &= \int_0^{\pi/2} \ln(\sin \theta) d\theta + \int_0^{\pi/2} \ln(\sin \theta) d\theta \\
 &= 2 \int_0^{\pi/2} \ln(\sin \theta) d\theta \quad |
 \end{aligned}$$

$$\begin{aligned}
 \text{let } I &= \int_0^{\pi/2} \ln(\sin \theta) d\theta \\
 &= \int_0^{\pi/2} \ln(\sin(\pi/2 - \theta)) d\theta \quad \text{using } \int_0^a f(x) dx = \int_0^a f(a-x) dx \\
 &= \int_0^{\pi/2} \ln(\cos \theta) d\theta
 \end{aligned}$$

$$\begin{aligned}
 \therefore 2I &= \int_0^{\pi/2} (\ln(\sin \theta) + \ln(\cos \theta)) d\theta \\
 &= \int_0^{\pi/2} \ln\left(\frac{1}{2} \sin 2\theta\right) d\theta \quad | \\
 &= \int_0^{\pi/2} (\ln(\sin 2\theta) - \ln 2) d\theta \\
 &= \int_0^{\pi/2} \ln(\sin 2\theta) d\theta - \int_0^{\pi/2} (\ln 2) d\theta \\
 &= \int_0^\pi \ln(\sin u) \times \frac{1}{2} du - \frac{\pi}{2} (\ln 2 - 0) \quad \text{let } u = 2\theta \\
 &\quad \quad \quad du = 2 d\theta
 \end{aligned}$$

$$2I = \frac{1}{2} \int_0^\pi \ln(\sin u) du - \frac{\pi}{2} \ln 2 \quad |$$

$$\begin{aligned}
 \therefore \int_0^\pi \ln(\sin \theta) d\theta &= 2I \\
 &= \frac{1}{2} \int_0^\pi \ln(\sin u) du - \frac{\pi}{2} \ln 2
 \end{aligned}$$

$$\begin{aligned}
 \therefore \frac{1}{2} \int_0^\pi \ln(\sin \theta) d\theta &= -\frac{\pi}{2} \ln 2 \quad | \\
 \int_0^\pi \ln(\sin \theta) d\theta &= -\pi \ln 2
 \end{aligned}$$